Specification and Modeling of Embedded System

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Theoretical Textbook

- Name: Introduction to embedded system（嵌入式系统）

  --A Cyber-Physical System Approach
- Author: Edward Ashford Lee
- Publish: UC berkeley , 2011

Reference book

- NAME: Principles of embedded Computation
- AUTHOR: Rajeev Alur
- PUBLISH: no published
- http://www.seas.upenn.edu/~cis540/

Logistics

- Class time : Tuesday 10:00-12:00

  Science Building：B218

- lab time: Monday 8:00-10:00(last 9 weeks)

  Science Building ：B226

Theoretical Content

- What this course is about
  - A principled, scientific approach to Modeling and Analyzing embedded systems
  - Focus on *model-based system design*, and on analysis of *embedded software*

Topics

- Introduction to Model-based Design
  - State machines
  - Dynamical systems （动力系统）
  - Timed Systems
  - Hybrid Systems （混成系统）
  - Observational Semantics（语义） and Refinement （精化）

Topics(cont’)

- Models of Interaction
  - Synchronous（同步） models
  - Asynchronous（异步） models
  - Dataflow networks
  - Time-triggered（时间触发） vs. event-triggered（事件触发） communication
  - Component-based Design

Topics(cont’)

- Specification（规范） and Verification（验证）
  - Requirements: Safety, Timely response, Liveness, Stability
  - Temporal logics （时态逻辑）
  - Analysis techniques: Simulation, Testing, Formal verification, and Monitoring
  - Basics of deductive verification: Invariants, Ranking functions （评级函数）
  - Model checking
  - Symbolic simulation
  - Abstractions and compositional verification

Experiment

- Tool: matlab
  - Base matlab
  - Stateflow
  - simulink